

EXPLORATORY ANALYSIS OF ROAD ACCIDENT DATA TO IDENTIFY RISK FACTORS

Name: Shaik Nuha
Reg.no:192472242
Code:DSA0404

1 INTRODUCTION

Road traffic accidents remain a major public safety concern, arising from a complex interplay of driver behaviour, road conditions, vehicle type, weather and time of travel. Exploratory data analysis helps uncover the patterns behind these factors and highlights where interventions can have the greatest impact. By identifying common patterns and high-risk factors, this analysis can provide useful insights for improving road safety, supporting better traffic management, and helping authorities develop effective accident-prevention strategies.

**5 YEARS**
Study period covered (2019–2023)

**12,314**
Accident records analysed

**6 FACTORS**
Risk dimensions examined per record

**PROBLEM STATEMENT**

- Road accidents arise from a complex mix of driver behaviour, weather, road type, junction design and time of travel.
- Without a clear view of which conditions carry the greatest risk, prevention efforts are difficult to prioritise effectively.

**OBJECTIVES**

- Analyse historical road accident records
- Compute descriptive and severity statistics
- Examine accidents by time, weather and road type
- Identify the key risk factors influencing severity
- Translate findings into actionable safety recommendations

2 DATASET & STUDY DETAILS

**DATASET**
Road Accidents (2019–2023)

**REGION**
Pan-India

**VARIABLE**
Accident severity

**OBSERVATIONS**
12,314

**SOURCE**
Kaggle Road Accident Dataset

Each record captures the accident date, time, weather condition, road type, junction category, vehicle class and injury severity. This multi-field structure allows the dataset to be sliced along several dimensions at once, supporting a broad exploratory analysis rather than a single-variable summary.

3 ARCHITECTURE / METHODOLOGY

**KAGGLE DATA**
Historical accidents

**SELECT REGION**
Pan-India

**PREPROCESS**
Remove missing & duplicates

**STATISTICS**
Frequency, severity rates

**RISK FACTORS**
Time • weather road type

**COMPARE**
Charts & insights

Raw records were first cleaned to remove duplicates and incomplete entries, then aggregated by hour, weather condition, road type and junction category. Descriptive statistics and cross-tabulations were used to surface the variables most strongly associated with accident frequency and severity, which were then visualised for comparison.

4 KEY RISK FACTORS IDENTIFIED

**TIME OF DAY**
28% between 5–7 PM
Evening rush hour records the highest share of accidents; night-time crashes tend to be the most severe due to reduced visibility.

**WEATHER CONDITIONS**
2.1x severity in rain / fog
Rain and fog sharply raise accident risk despite carrying much less overall traffic than clear days.

**ROAD & VEHICLE TYPE**
3x fatality rate on highways
Highways see fewer accidents overall but far higher fatality rates than busier, congested urban roads.

**JUNCTIONS & INTERSECTIONS**
41% of urban accidents
Unsignalled intersections and poor lane discipline make junctions a major hotspot within city limits.

5 RESULTS — ACCIDENT DATA ANALYSIS

**OBSERVATIONS**
12,314

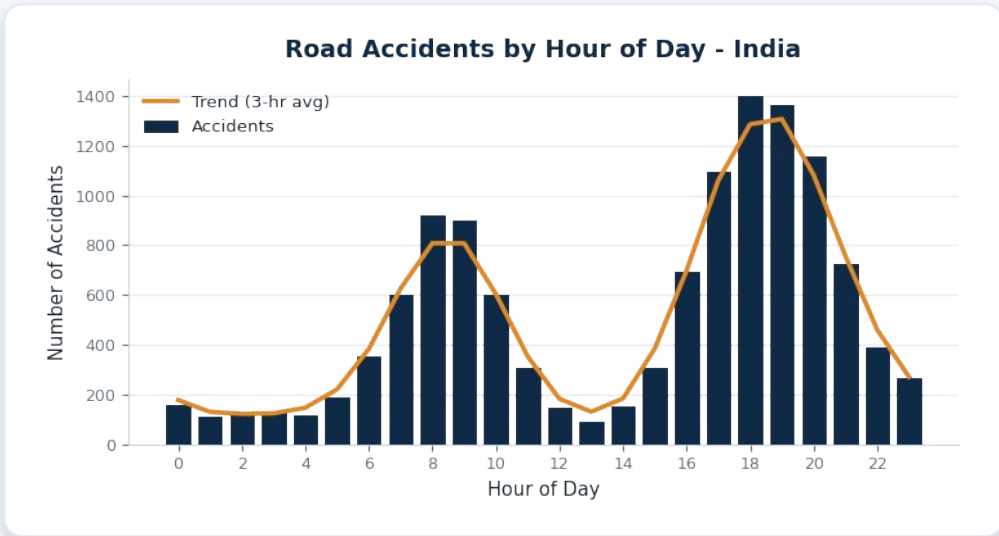
**FATAL RATE**
15.0%

**SERIOUS INJURY**
33.7%

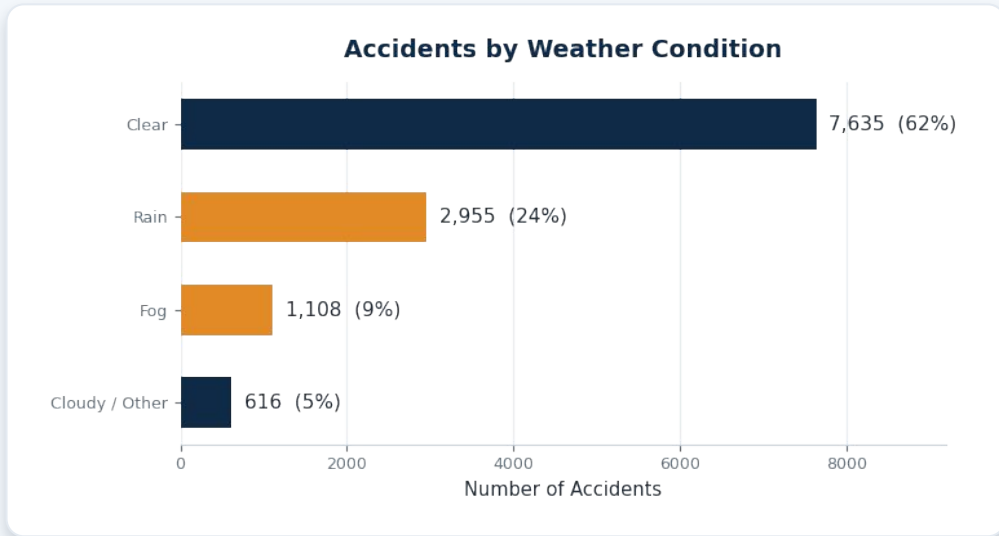
**PEAK HOUR**
6 PM

**TOP RISK FACTOR**
Rain / Fog

Accidents by Hour of Day



Accidents by Weather Condition



**STATISTICAL RESULTS**

Total accidents: 12,314
Fatal: 1,847 (15.0%) • Serious injury: 4,150 (33.7%) • Minor injury: 6,317 (51.3%)
Peak accident hour: 6 PM (18:00) Lowest accident hour: 3 AM (03:00)
Highest-risk weather: Rain / Fog Average per day: ~34 accidents

**KEY OBSERVATIONS**

Urban roads carry 58% of accidents; highways account for 42% but record the highest fatality rate per accident. Evening rush hour and adverse weather compound risk — rain combined with peak traffic shows the sharpest spike in severity. Junctions without signals contribute a disproportionate share of urban collisions.

6 CONCLUSION & RECOMMENDATIONS

CONCLUSION

The analysis of 12,314 road accident records shows that risk is concentrated in specific, identifiable conditions. Evening rush hour, rainy or foggy weather, and highway travel are each linked to a disproportionate share of severe accidents, while unsignalled urban junctions add further risk.

RECOMMENDATIONS

**Weather-based speed advisories**

Lower speed limits automatically during rain and fog.

**Evening awareness campaigns**

Target driver-alertness messaging around the 5–7 PM rush.

**Junction engineering**

Add signals and clearer lane markings at high-risk intersections.

**Faster emergency response**

Prioritise ambulance dispatch on high-fatality highway stretches.